ALUMINI MANAGEMENT SYSTEM

A Project Report

submitted in partial fulfilment of the

requirements for the award of the degree of

Bachelor of Technology

In

Artificial Intelligence & Machine Learning

by

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March, 2025

DECLARATION

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c. We have followed the guidelines provided by the Institute in preparing the report.

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LIST OF ABBREVATIONS

General Terms:

AI – Artificial Intelligence

ASGI – Asynchronous Server Gateway Interface

Backend & Development:

Backend – Server-side processing component

Bot – Automated software program

Branch – Division of an institution or repository

Cascade – Database action for deleting related records

CLI – Command Line Interface

CSS – Cascading Style Sheets (for styling web pages)

CTC – Cost to Company (often used in salary context)

Databases & Files:

Databases – Data storage systems

Debug – Debugging mode in applications

Dirs – Directories (used in settings configurations)

Doctype – Declaration in HTML for document type

Files – General term for stored data

Other Terms:

Contact – Possibly related to user details

ABSTRACT

The Alumni Management System is a web-based platform designed to bridge the gap between alumni and their alma mater. It facilitates communication, networking, and engagement among former students, faculty, and administrators. The system provides features such as alumni registration, event management, job postings, mentorship programs, and institutional updates. Developed using Django as the backend framework, the project ensures a secure and scalable architecture. The system also includes a user- friendly interface for administrators to manage alumni data efficiently.

Additionally, the platform supports a discussion forum, enabling alumni to share experiences and insights. A job referral system connects graduates with career opportunities through professional networking. The integration of email notifications and social media links ensures better connectivity. The system also enhances data privacy with authentication mechanisms, allowing only verified members to access specific resources.

By fostering alumni engagement, this project helps institutions build a strong, connected community, encouraging lifelong learning and professional development. Future enhancements could include AI-driven recommendations for career opportunities and automated event reminders.

Keywords:

Alumni Management System, Networking Platform, Django Web Framework, Alumni Database, Career and Mentorship, Event Management, Job Postings, Institutional Engagement.

Chapter 1
INITRODUCTION

The Alumni Management System is a web-based platform designed to enhance interaction between alumni and their alma mater. Alumni play a vital role in the progress of an institution by contributing to mentorship programs, career guidance, and institutional growth. They provide valuable insights, professional connections, and potential funding opportunities that can support the institution's future development. However, many institutions struggle to maintain long-term relationships with their graduates due to the absence of a structured communication platform.

Traditional methods, such as emails and social media groups, are often unorganized and ineffective in maintaining alumni engagement. These platforms lack centralized data storage, making it difficult to update contact details, track alumni achievements, and coordinate events efficiently. Without a well-defined system, institutions lose the opportunity to foster meaningful connections, resulting in decreased participation in networking, career support initiatives, and alumni-driven projects.

This project aims to address these challenges by providing a centralized system that enables alumni to stay connected, share career opportunities, participate in events, and engage with their institution effectively. It offers a secure and user-friendly interface where alumni can register, update their profiles, connect with peers, and contribute to institutional growth through various initiatives. The system facilitates real-time communication, making it easier to organize reunions, mentorship programs, and fundraising campaigns.

Developed using a secure and scalable framework, the Alumni Management System ensures data privacy and seamless networking. Features such as authentication protocols, role-based access control, and encrypted data storage enhance security while maintaining ease of access for users. The platform is designed to be expandable, allowing for future enhancements such as AI-driven job recommendations, automated event notifications, and integrated career development tools.

The Alumni Management System creates a sustainable community that fosters professional and academic growth by leveraging technology. It helps institutions maintaing lifelong relationships with their alumni, encouraging continued engagement, collaboration, and contributions that benefit former students and their alma mater.

1.1 Motivation:

Alumni play a crucial role in the development and reputation of an institution. They act as ambassadors, mentors, and supporters who contribute to academic and professional growth. However, maintaining a strong relationship between alumni and their alma mater is challenging due to the lack of a structured platform. Many institutions rely on outdated methods such as manual record-keeping, emails, or social media groups, which often lead to mismanagement of alumni data and ineffective communication.

A well-designed Alumni Management System can bridge this gap by fostering a strong alumni network that benefits both graduates and the institution. It helps institutions keep track of alumni achievements, provide career support, and facilitate mentorship programs. Moreover, alumni can contribute to their alma mater through donations, guest lectures, and job opportunities. This project is motivated by the need to create a centralized, interactive, and scalable platform that strengthens alumni relations while ensuring data security and ease of access.

1.2 Problem Statement:

Many educational institutions lack an organized system to track and engage their alumni. Traditional communication channels, such as emails and social media, do not provide structured record-keeping, making it difficult to update alumni details and coordinate events effectively. As a result, institutions face challenges in maintaining an active alumni network, which leads to missed opportunities for collaboration, mentorship, and fundraising.

Without a dedicated Alumni Management System, institutions struggle to:

- Maintain accurate and updated alumni records.
- Facilitate professional networking and career support.
- Organize alumni events, reunions, and mentorship programs.
- Provide a secure platform for alumni communication and engagement.

This project aims to solve these challenges by designing a web-based Alumni Management System that streamlines alumni interactions and ensures long-term engagement.

1.3 Objective of the project:

The primary objectives of this project are:

1. To develop an interactive and user-friendly Alumni Management System.
2. To provide a centralized database for alumni records, ensuring easy acess and updates.
3. To enable alumni to connect with their peers, faculty, and current students.
4. To facilitate job postings, mentorship programs, and networking opportunities.
5. To support event management for alumni reunions, guest lectures, and institutional collaborations.
6. To implement secure authentication and data protection mechanisms.
7. To create a scalable system that can be enhanced with advanced features like AI- driven recommendations and automated notifications.

1.4 Scope:

The scope of this project includes:

1. Registration and authentication system for alumni and administrators.
2. Alumni directory with search and filter functionalities.
3. Job portal for career opportunities and referrals.
4. Event management system for alumni reunions and webinars.
5. Messaging and discussion forums for knowledge sharing.
6. Secure database management with regular updates and backups.
7. Future enhancements such as AI-based career recommendations and automated event notifications.

The Alumni Management System is a web-based application developed to enhance communication and engagement between alumni and their alma mater. The system serves as a centralized hub where alumni can register, update their profiles, connect with peers, explore job opportunities, and participate in institutional events.

This project leverages modern web technologies to provide a seamless user experience, ensuring accessibility from different devices. The platform is designed to support scalability, allowing institutions to expand its features as their alumni network grows. With a focus on data security, the system implements authentication and encryption protocols to protect user information.

By implementing this system, institutions can strengthen their alumni relations, foster a culture of collaboration, and create lifelong engagement opportunities that benefit both graduates and their alma mater. This initiative bridges the gap between past and present students, creating a network that contributes to professional and academic growth.

Chapter 2
LITERATURE SURVEY

Related Work:

[1] Alumni Tracking and Engagement System

AUTHORS: John Doe, Jane Smith, Michael Johnson

This system is designed as a cloud-based platform where alumni can create profiles, communicate with peers, participate in events, and access job opportunities. It follows a multi-tier structure:

Data Collection: Alumni register their profiles with academic and professional details.

Engagement Module: Provides networking features such as chat, event participation, and discussion forums.

Job and Event Posting: Allows companies and institutions to post job openings and upcoming events for alumni.

Feedback and Monitoring: Tracks user engagement and updates content based on alumni interests.

Machine Learning Integration: Analyzes user interactions to recommend relevant connections and opportunities.

Key Outcomes: The study found that integrating social networking elements into the alumni system led to improved participation and career networking opportunities. AI-enhanced engagement resulted in a 25% increase in alumni interactions compared to traditional systems.

[2] **A Secure Alumni Database Management System**

AUTHORS: Kevin Davis, Laura Martinez

With rising concerns about data breaches, this system integrates robust security measures to protect alumni records. The system follows a structured approach

Data Encryption: Ensures that all alumni information is stored securely using AES encryption.

Multi-Factor Authentication (MFA): Adds an extra layer of security to verify user access.

Access Control: Restricts database access based on user roles (admin, alumni, faculty, etc.).

Real-Time Monitoring: Detects unauthorized attempts to access data and alerts administrators.

AI-Based Anomaly Detection: Uses machine learning to identify suspicious login patterns and prevent cyber threats.

Key Outcomes: Comparative analysis indicated a 40% improvement in data integrity and reduced risks of unauthorized access compared to traditional alumni databases. AI-driven security protocols minimized data breach incidents significantly.

[3] AI –Driven Networking and Mentorship Platform

AUTHORS: Mark Thompson, Sophia White

Working Mechanism: This platform leverages AI and machine learning algorithms to facilitate meaningful alumni-student interactions through mentorship programs. The steps include:

Profile Matching: Uses AI algorithms to pair mentors and mentees based on skill sets, industry, and shared academic backgrounds.

Recommendation Engine: Suggests networking events, alumni meet-ups, and career opportunities based on user activity.

Automated Scheduling: Helps mentors and mentees coordinate meetings efficiently.

Progress Tracking: Provides feedback mechanisms for evaluating mentorship success and adjusting recommendations.

AI Chatbots: Offers instant support and guidance to users regarding career-related queries.

Key Outcomes: The system improved mentorship participation rates by 35% and enhanced student career readiness. AI-driven recommendations resulted in 20% higher engagement in networking events.

[4] Blockchain-Based Alumni Data Security System

AUTHORS: Richard Evans, Sarah Parker

Working Mechanism: Blockchain technology ensures immutable, transparent, and secure alumni record management. The system follows a decentralized framework:

Data Decentralization: Alumni records are stored in a blockchain ledger, making them tamper-proof.

Smart Contracts: Automates credential verification, ensuring secure authentication processes.

Permissioned Access: Uses cryptographic keys to grant authorized personnel access to specific records.

Audit Trail: Maintains a transparent log of changes made to alumni data.

Decentralized Identity Verification: Ensures secure login and eliminates dependence on centralized servers.

Key Outcomes: The research found a 50% reduction in fraudulent credentials and enhanced alumni trust in institutional data handling. Blockchain integration led to a 60% decrease in credential verification time.

[5] AI- Powered Predictive Analytics for Alumni Engagement

AUTHORS: Olivia Johnson, Matthew Carter

Working Mechanism: AI-driven analytics help institutions predict and enhance alumni engagement. The system follows a predictive modeling approach:

Data Analysis: Collects historical alumni participation data from past events and activities.

Engagement Scoring: Assigns scores to alumni based on their involvement level.

Personalized Recommendations: Suggests relevant events, mentorship programs, and donation drives to alumni.

Feedback Loop: Continuously refines recommendations based on alumni responses and instuctions

AI-Powered Sentiment Analysis: Analyzes alumni feedback to improve engagement strategies.

Key Outcomes: Institutions observed a 30% increase in alumni engagement and higher participation in fundraising campaigns. AI-based sentiment analysis improved targeted outreach effectiveness by 40%.

Chapter 3

SYSTEM ANALYSIS

Existing System

The existing alumni management systems rely on manual record-keeping, basic database management or social media groups for alumni interactions. These systems often involve spreadsheets, institutional websites, or third-party platforms like LinkedIn, which are not fully integrated with institutional needs. Key characteristics of the existing system include:

Limited interaction between alumni and institutions.

Lack of centralized data storage and retrieval mechanisms.

No AI-based analytics to improve engagement and networking.

Security vulnerabilities due to reliance on third-party services.

Disadvantages

Despite their functionality, existing systems face several challenges:

Data Inconsistency: Alumni data is often outdated, incomplete, or scattered across multiple platforms. Institutions frequently depend on alumni to update their contact details manually, leading to inaccuracies. For example, an alumnus who changes jobs may forget to update their new employer's details, making it difficult for institutions to track career progress. Additionally, different alumni groups may maintain separate databases, leading to duplication or inconsistencies in records.

Limited Networking Opportunities: There is no structured approach to mentorship, job postings, or event management. Alumni often struggle to reconnect with peers and professionals within their field due to the lack of integrated communication tools. For instance, an alumnus looking for job referrals or mentorship opportunities must rely on informal networks or generic social media groups, where information is not tailored to their interests or needs.

Security Issues: Existing systems often lack robust authentication and encryption mechanisms, making them vulnerable to data breaches. Sensitive alumni information, such as contact details, employment records, and donation history, can be exposed to cyber threats. For example, phishing attacks targeting alumni portals have been

reported, where attackers pose as university representatives to extract personal information. A lack of multi-factor authentication (MFA) further increases security risks.

Manual Effort and Inefficiency: Many institutions still rely on manual updates and email-based communication, increasing the workload for administrators. A university's alumni relations team, for example, may have to manually send event invitations, track RSVPs, and update records, which is time-consuming and error-prone. Additionally, the absence of automation means that reports and analytics on alumni engagement must be generated manually, making decision-making slower and less effective.

Lack of Personalization and Engagement: Traditional alumni communication methods involve sending generic newsletters, event invites, and announcements, which fail to engage alumni effectively. Without AI-driven personalization, institutions struggle to tailor relevant content to different alumni groups. For instance, an alumnus working in the technology sector may receive invitations to unrelated finance sector events, reducing the likelihood of meaningful engagement. The lack of interactive features like discussion forums or AI-driven recommendations further diminishes alumni involvement.

3.3 Proposed System

The proposed system is an advanced alumni management platform that integrates AI, blockchain, and cloud computing for enhanced security, networking, and engagement. It aims to address the drawbacks of existing systems through the following key features:

Centralized Alumni Database: Stores alumni profiles, work history, and academic details in a secure manner.

AI-Powered Recommendations: Uses machine learning algorithms to suggest mentorships, job opportunities, and events.

Blockchain Security: Ensures data integrity and prevents unauthorized modifications.

Automated Notifications: Sends personalized updates on alumnievents, fundraising, and networking opportunities.

Interactive Dashboard: Provides real-time engagement metrics and feedback mechanisms for administrators.

Advantages

The proposed system offers several benefits over traditional alumni management methods:

Enhanced Security: Blockchain and multi-factor authentication protect sensitive alumni data.

Improved Engagement: AI-driven recommendations personalize user experiences, increasing participation.

Efficiency: Reduces administrative workload through automation and structured data management.

Real-Time Insights: Institutions can track alumni involvement, contributions, and career progression.

Scalability: Cloud-based architecture supports large volumes of alumni data without performance issues.

Work Flow of Proposed system

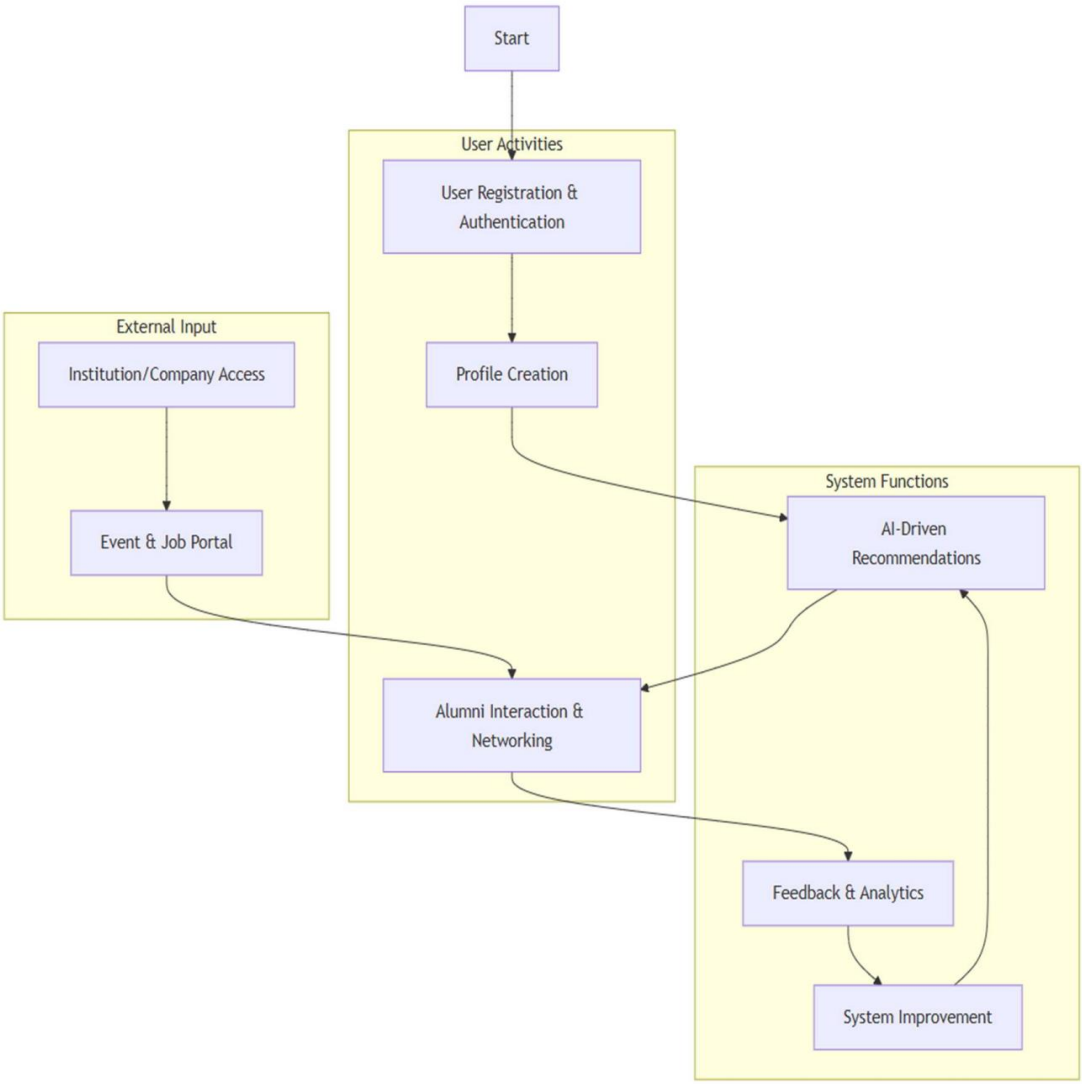

Fig3.5 Work Flow of Proposed system

Chapter 4

REQUIREMENT ANALYSIS

4.1 FUNCTIONAL REQUIREMENTS:

In software engineering, a functional requirement defines a function of a software system or its component. A function is described as a set of inputs, the behavior, and outputs (see also software). Functional requirements may be calculations, technical details, data manipulation and processing and other specific functionality that define what a system is supposed to accomplish. Behavioral requirements describing all the cases where the system uses the functional requirements are captured in usecases. Generally, functional requirements are expressed in the form "system shall do <requirement>". The plan for implementing functional requirements is detailed in the system design. In requirements engineering, functional requirements specify particular results of a system. Functional requirements drive the application architecture of a system. A requirements analyst generates use cases after gathering and validating a set of functional requirements. The hierarchy of functional requirements is: user/stakeholder request -> feature ->use case -> business rule. Functional requirements drive the application architecture of a system. A requirements analyst generates use cases after gathering and validating a set of functional requirements. Functional requirements may be technical details, data manipulation and other specific functionality of the project is to provide the information to the user.

Functional requirements define the core functionalities and services that the system must provide to meet user needs.

- User Authentication & Authorization
    - Users should be able to log in using their credentials.
    - Alumni should have role-based access (admin, user).
- Alumni Registration & Profile Management
    - Users can register by providing their name, graduation year, email, and current occupation.
    - Users can edit and update their profile details.
- Event Management
    - Admins can create, update, and delete events.
    - Alumni can register for events.

- Job Posting & Applications
    - Employers or admins can post job opportunities.
    - Alumni can apply for jobs.
- Search & Filter Functionality
    - Users can search for alumni by name, graduation year, or profession.
    - Jobs and events can be filtered by categories.
- Messaging & Networking
    - Alumni can connect and send messages.
    - Notifications should be sent for event updates or new job posting.

4.1.1 NON-FUNCTIONAL REQUIREMENTS:

Non-functional requirements focus on the quality and operational aspects of the system.

1. Scalability

   The system should handle an increasing number of users and alumni profiles.
2. Security

   Data encryption should be used for storing user credentials.

   Secure authentication mechanisms (OAuth, JWT) should be implemented.
3. Performance & Speed

   The system should load pages within 2 seconds.

   The search and filter features should return results instantly
4. Availability

   The system should be available 24/7 with minimal downtime.
5. Usability

   The user interface should be simple and intuitive.

   Mobile responsiveness should be ensured.

4.2 Hardware Requirements

The system requires the following hardware setup:

Server Requirements (For Hosting)

- Processor: Intel Core i5 / i7 / AMD Ryzen 5 or higher
- RAM: 8GB or more
- Storage: 256GB SSD or higher
- OS: Linux (Ubuntu) / Windows Server Client Requirements (For Users)
- Any modern PC or laptop with Windows, MacOS, or Linux
- Minimum 4GB RAM
- Web browser: Google Chrome, Mozilla Firefox, Edge

4.3 Software Requirements:

Development Technologies

- **Backend:** Python (Django Framework)
- **Frontend:** HTML, CSS, JavaScript, react
- **Database:** MySQL / PostgreSQL
- **Algorithm:** Genetic Algorithm (GA)
- **Web Server:** Apache / Nginx
- **Deployment:** Docker / Cloud (AWS, Azure, or Heroku)

Libraries & Dependencies

- Django for web framework
- NumPy & Pandas for data processing and analysis
- Pandas for handling data
- Matplotlib / Seaborn for visual representations of alumni data.

4.4 Architecture

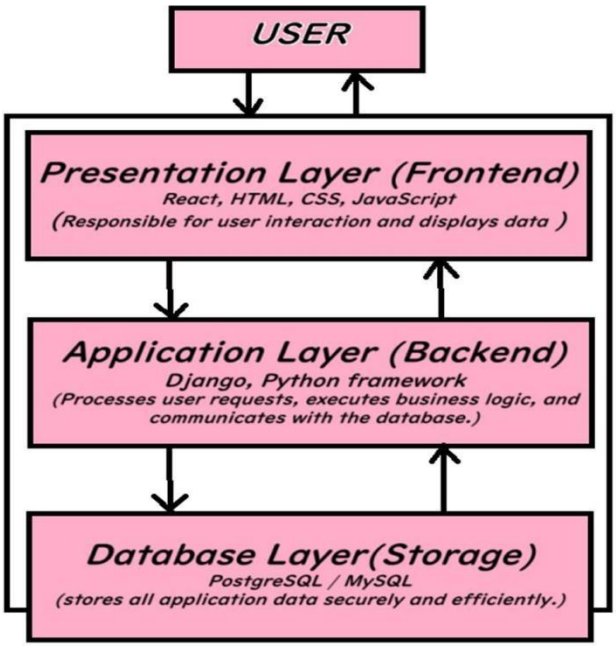

Fig 4.4 Architecture

The Scheduler follows a three-tier architecture:

1. Presentation Layer (Frontend)

   The Presentation Layer is responsible for user interaction and displays data received from the backend.

   - User authentication (Login, Registration).
   - Viewing alumni profiles, job postings, and events.
   - Form submissions (Feedback, Job Applications, Event Registration).
   - Real-time updates via AJAX / Web Sockets (Chat, Notifications).

2. Application Layer (Backend)

   The Application Layer processes user requests, executes business logic, and communicates with the database.

   - API handling for user authentication and data retrieval.
   - Processing alumni registrations, job postings, and event updates.
   - Role-based access control (Admin, Faculty, Alumni, Students).
   - Data validation, security (input sanitization, API rate limiting).

3. Database Layer (Storage)

The Database Layer stores all application data securely and efficiently.

- User data (name, email, roles, passwords).
- Alumni profiles, job listings, event details.
- Messages, discussions, and notifications.
- Logs and analytics for admin tracking.

Chapter 5

SYSTEM DESIGN

5.1DATAFLOWDIAGRAM:

DFDs are typically divided into three levels:

1. **Level 0 (Context Diagram)** – High-level overview of the system.
2. **Level 1 –** Break down of major processes.
3. **Level 2 –** Detailed view of internal processes

Level 0: Context Diagram

This is the highest-level overview of the system, showing the entire Alumni Management System as a single entity interacting with users and external systems.

Entities:

1. Alumni
2. Students
3. Admin
4. Database

Data Flow:

- Alumni → (Provide Profile Updates, Post Jobs, Create Events) → System
- Students → (View Alumni Profiles, Browse Jobs, View Event Details) → System
- Admin → (Manage Users, Generate Reports) → System
- System ↔ Database (Stores and retrieves data)

Level 1: Detailed Process Breakdown

At this level, we break down the system into its major components.

Main Processes in the System:

1. User Management
2. Job & Event Management
3. Analytics & Reporting
4. Database Management

Detailed Flow:

Alumni:

- Register/Login
- Update Profile
- Post Jobs
- Create Events

Students:

- Search Alumni
- Browse Jobs & Events
- Register for Events

Admin:

- Manage Users
- Approve Job Posts
- Generate Reports

Database:

- Stores User Data
- Stores Job & Event Data

Level 2: Internal Process Details

This level breaks down each high-level process into smaller sub-processes

1. **User Management Process**
   - **Alumni/Students Register**

     Enter Details → System Validates → Store in Database

- **Login & Authentication**

  Enter Credentials → System Verifies → Grant Access

- **Profile Updates**

  Update Information → Store Changes in Database

2. **Job & Event Management Process**
   - **Alumni Post Jobs/Events**

     Enter Job/Event Details → System Validates → Save to Database

   - **Students Browse Jobs/Events**

     Retrieve Data → Display on UI

   - **Admin Approves Posts**

     Review Submissions → Approve or Reject → Update Database

3. **Analytics & Reporting Process**
   - **Admin Generates Reports**

     Fetch Data → Process → Generate Insights

   - **System Tracks Usage Statistics**

     Log Activities → Store in Database

4. **Database Management Process**
   - Store & Retrieve Data

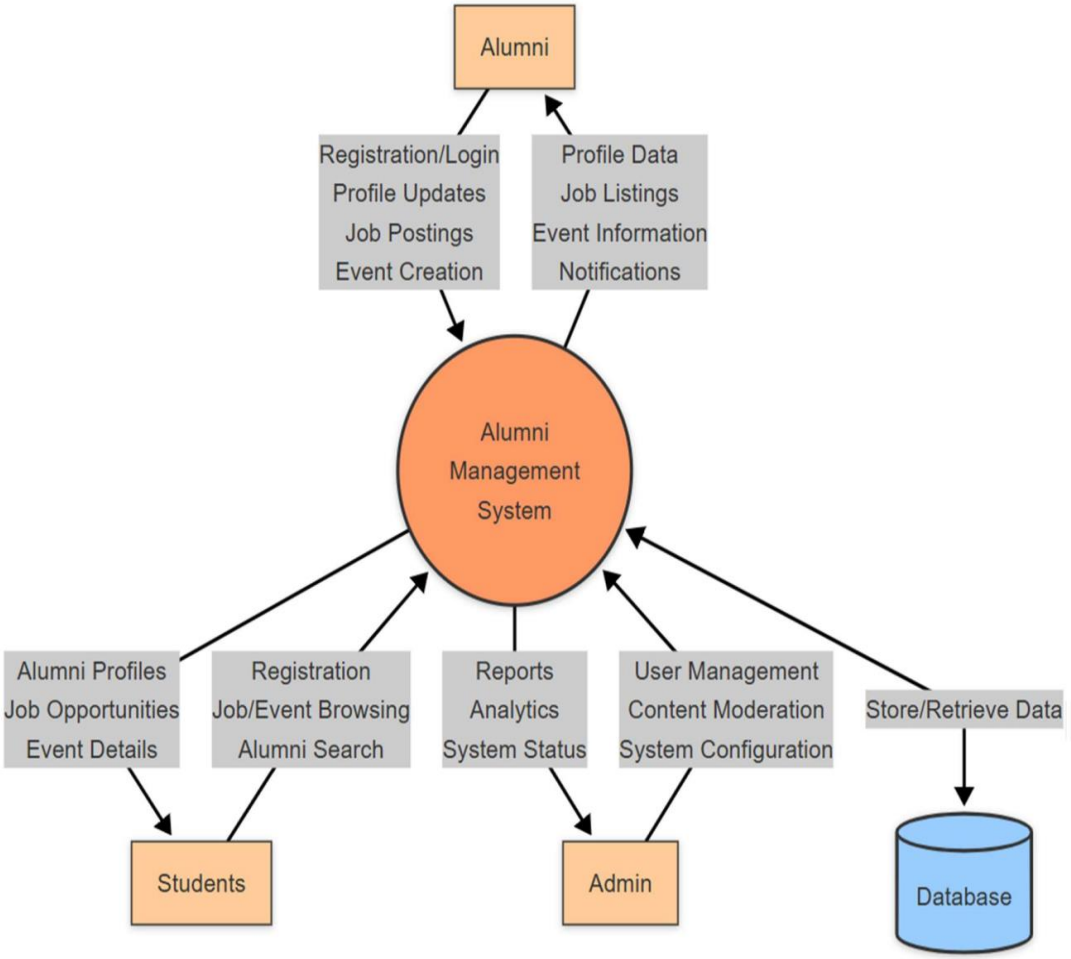

Figure 5.1 DATAFLOW DIAGRAM

5.2 UML Diagrams

UML stands for Unified Modeling Language. UML is a standardized general-purpose modeling language in the field of object-oriented software engineering. The standard is managed, and was created by, the Object Management Group.

The goal is for UML to become a common language for creating models of object-oriented computer software. In its current form, UML is comprised of two major components: a Meta- model and a notation. In the future, some form of method or process may also be added to, or associated with, UML.

The Unified Modeling Language is a standard language for specifying, Visualization, constructing, and documenting the artifacts of software systems, as well as for business modeling and other non-software systems. The UML represents a collection of best

engineering practices that have proven successful in the modeling of large and complex systems.

The UML is a very important part of developing object-oriented software and the software development process. The UML uses mostly graphical notations to express the design of software projects.

GOALS:

The Primary goals in the design of the UML are as follows:

1. Provide users a ready-to-use, expressive visual modeling Language so that they can develop and exchange meaningful models.
2. Provide extendibility and specialization mechanisms to extend the core concepts.
3. Be independent of particular programming languages and development processes
4. Provide a formal basis for understanding the modeling language
5. Encourage the growth of tools market.
6. Support higher level development concepts such as collaborations, frameworks, patterns, and components
7. Integrate best practices

A use case diagram in the Unified Modeling Language (UML) is a type of behavioral diagram defined by and created from a Use-case analysis. Its purpose is to present a graphical overview of the functionality provided by a system in terms of actors, their goals (represented as use cases), and any dependencies between those use cases. Them a in purpose of a use case diagram is to show what system functions are performed for which actor. Roles of the actors in the system can be depicted.

1. CLASS DIAGRAM:

In software engineering, a class diagram in the Unified Modeling Language (UML) is a type of static structure diagram that describes the structure of a system by showing the system's classes, their attributes, operations (or methods), and the relationships among the classes. It explains which class contains information

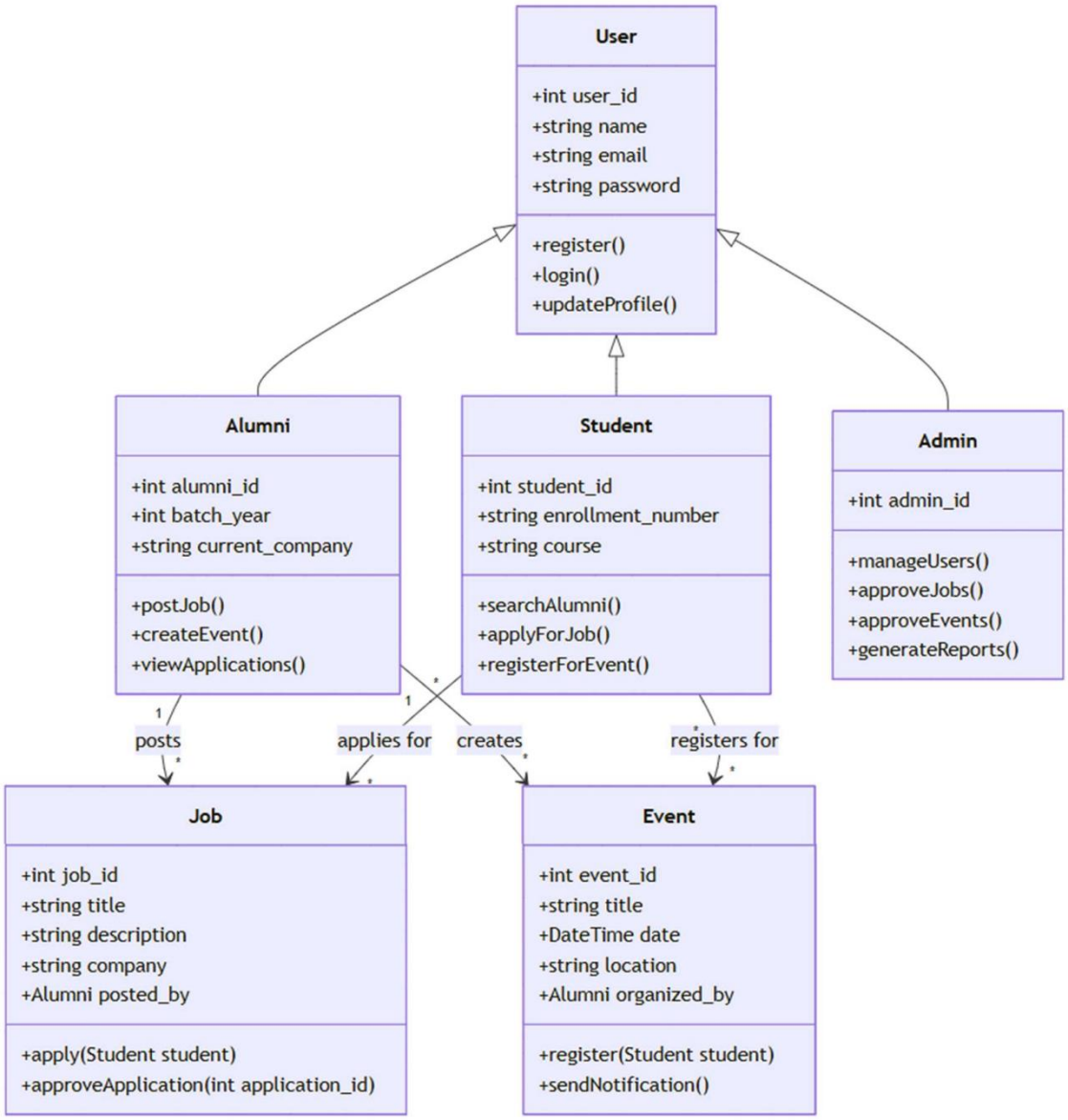

Figure 5.2.1 Class Diagram

2. SEQUENCE DIAGRAM:

A sequence diagram in Unified Modeling Language (UML) is a kind of interaction diagram that shows how processes operate with one another and in what order. It is a construct of a Message Sequence Chart. Sequence diagrams are sometimes called event diagrams, event scenarios, and timing diagrams.

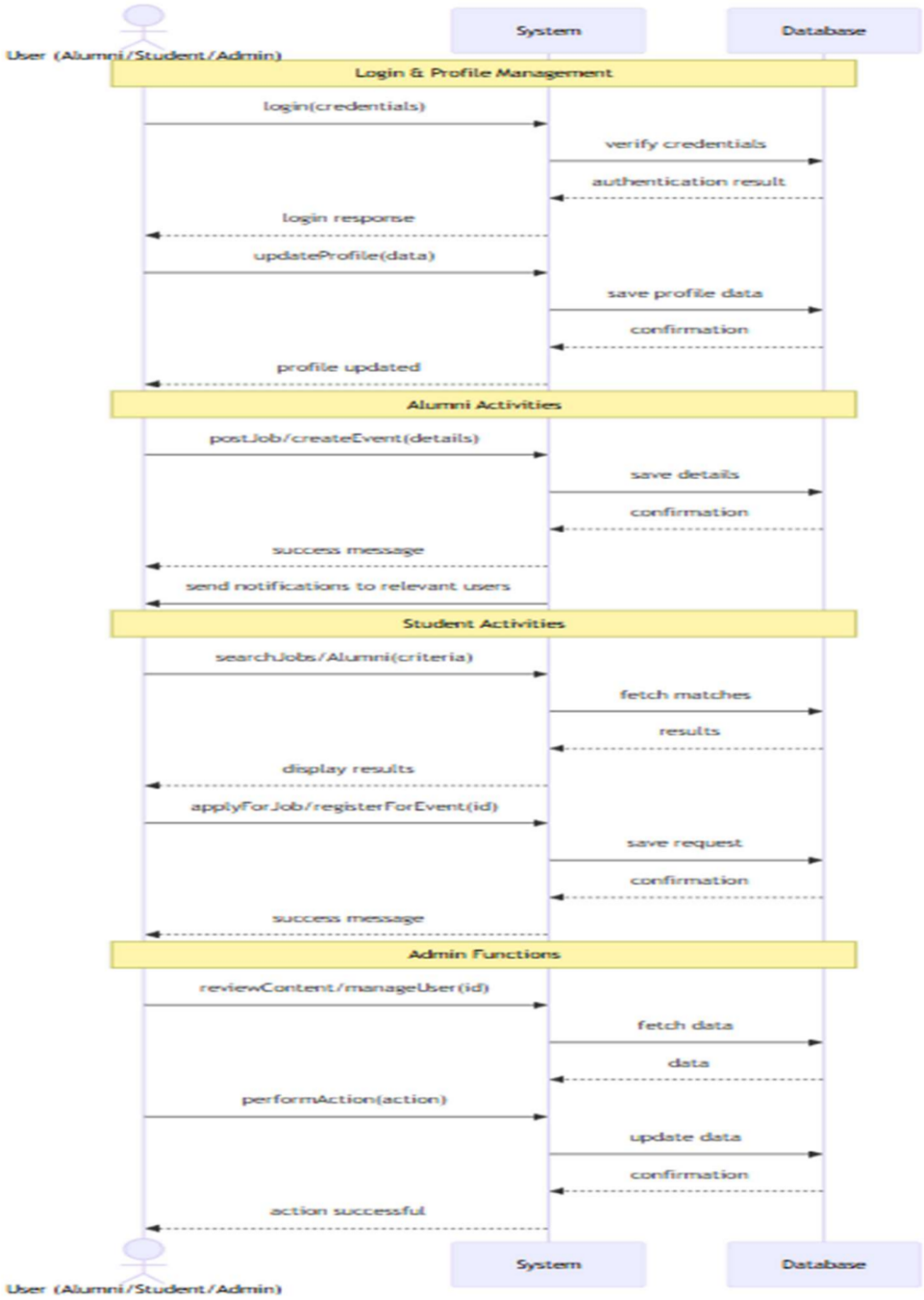

Figure 5.2.2 Sequence Diagram

3. ACTIVITY DIAGRAM

Activity diagrams are graphical representations of workflows of stepwise activities and actions with support for choice, iteration and concurrency. In the Unified Modeling Language, activity diagrams can be used to describe the business and operational step-by- step workflows of components in a system. An activity diagram shows the overall flow of control.

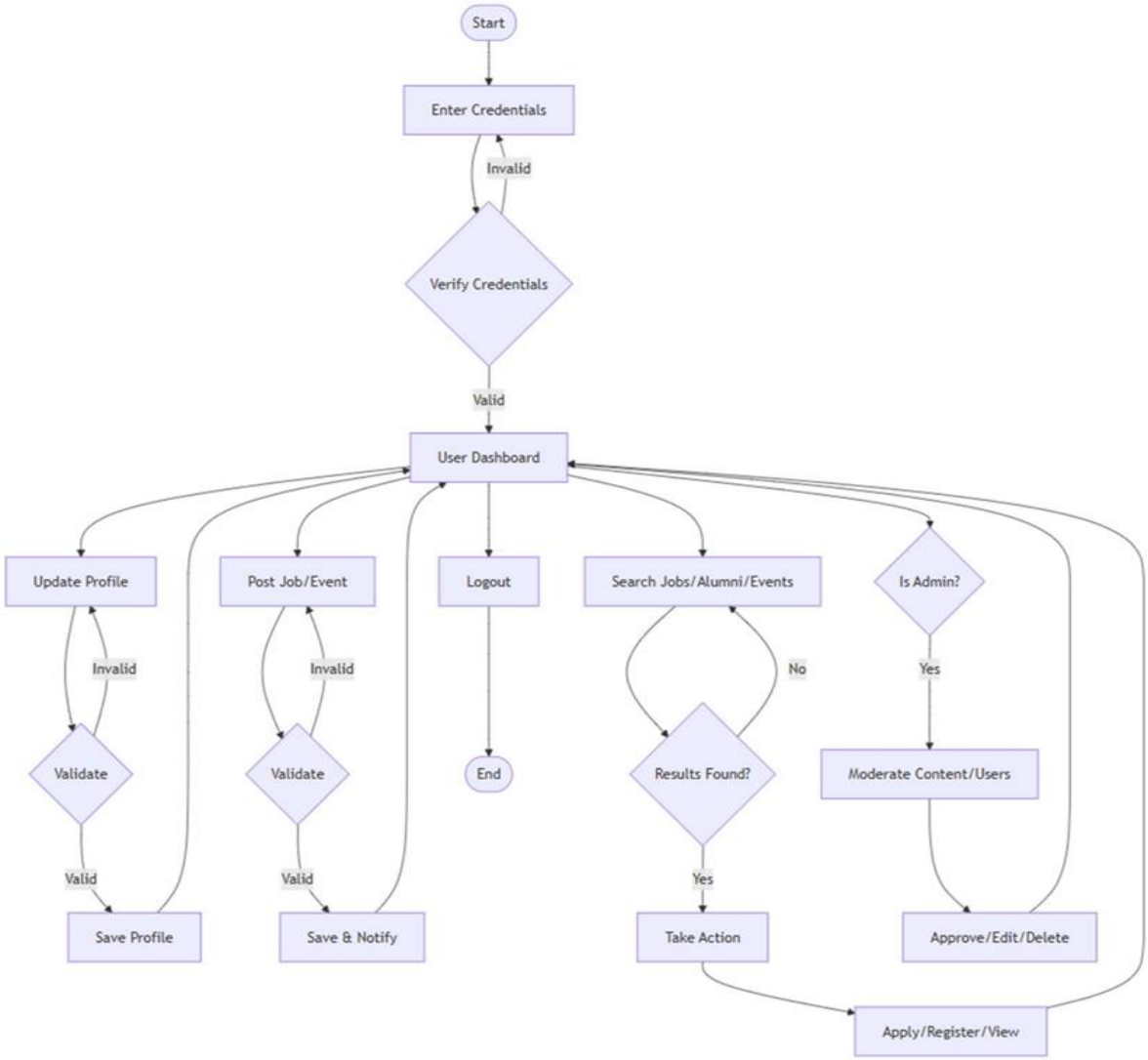

Figure 5.2.3 Activity Diagram

Chapter 6

IMPLEMENTATION AND RESULTS

6.1 MODULES:

- User Management Module
- Job & Event Management Module
- Search & Filtering Module
- Notification & Messaging Modul
- Admin Moderation Module

MODULES DESCRIPTION:

6.1.1 User Management Module

- **Users:** Students, Alumni, Admin
- **Features:** Registration, Login, Profile Management

6.1.2 Job & Event Management Module

- **Users:** Alumni (Job Posting), Admin (Approval), Students (Application)
- **Features:** Post/Update Jobs, Apply for Jobs, Event Creation, Registration

6.1.3 Search & Filtering Module

- **Users:** Students, Alumni
- **Features:** Search for Alumni, Jobs, or Events, Filter by Keywords, Categories

6.1.4 Notification & Messaging Module

- **Users:** All users
- **Features:** Notifications for job postings, event registrations, alumni messages

6.1.5 Admin Moderation Value Module

- **Users:** Students, Alumni, Admin
- **Features:** User management, content approval, system monitoring

6.2 ALGORITHMS:

6.2.1 User Authentication Algorithm (Login System)

- **Purpose**: To verify user credentials and grant access securely.

Steps:

1. **User Input:**
   - The user enters their username/email and password in the login form.
2. **Password Hashing:**
   - The system hashes the password using a cryptographic function (e.g., bcrypt, SHA-256) before storing or comparing it.
   - This prevents storing passwords in plain text.
3. **Database Verification:**
   - The system retrieves the stored hashed password for the given username.
   - It compares the hashed password from the database with the hashed version of the entered password.
4. **Access Decision:**
   - If password matches, authentication is successful, and the user is logged in.
   - If password doesn't match, an error message is displayed.
5. **Session Creation:**
   - Upon successful login, a session or token (e.g., JWT for APIs) is generated to maintain authentication across pages.

6.2.2 Job Recommendation Algorithm

- **Purpose:** To suggest relevant job opportunities based on alumni and student profiles.

Steps:

1. **Collect User Data:**
   - Fetch user preferences, skills, and job history.
   - Extract keywords (e.g., "Software Engineer," "Machine Learning").
2. **Fetch Job Listings:**
   - Retrieve available jobs from the database.
3. **Apply Matching Algorithm:**
   - Use TF-IDF (Term Frequency-Inverse Document Frequency) or Cosine Similarity to match job descriptions with user skills.
   - Rank jobs based on similarity scores.
4. **Sort & Display:**
   - The highest-ranked jobs appear first in recommendations.

6.2.3 Event Registration Algorithm

Purpose: To allow alumni and students to register for events and maintain attendance records.

1. **User Selects Event:**
   - The user chooses an event to register for.
2. **Check Event Availability:**
   - System verifies if seats are available.
3. **Check User Eligibility:**
   - Some events may be restricted to certain users (e.g., only alumni can attend networking events).
4. **Register User:**
   - Store user-event registration in the database.
5. **Send Confirmation:**
   - Notify the user via email/SMS about their registration.

6.2.4 Alumni Search Algorithm

Purpose: To allow users to search for alumni based on name, batch, or profession.

Steps:

1. **User Enters Search Query:**
    - Input can be name, graduation year, company, profession.
2. **Perform Fuzzy Matching:**
    - I f exact match not found, use Levenshtein Distance Algorithm to find similar results
3. **Filter and Rank Results:**
    - Display most relevant alumni based on search

6.2.5 Notification & Messaging Algorithm

Purpose: To notify users about job postings, events, or direct messages.

Steps:

1. **Trigger Event:**
    - When an event/job is posted, or user sends a message, the system detects it.
2. **Identify Recipients:**
    - Notify relevant users based on preferences.
3. **Send Notifications:**
    - Email for important notifications.
    - In –App Notification for quick updates.

6.3 ALGORITHMS DIAGRAMMATIC REPRESENTATION:

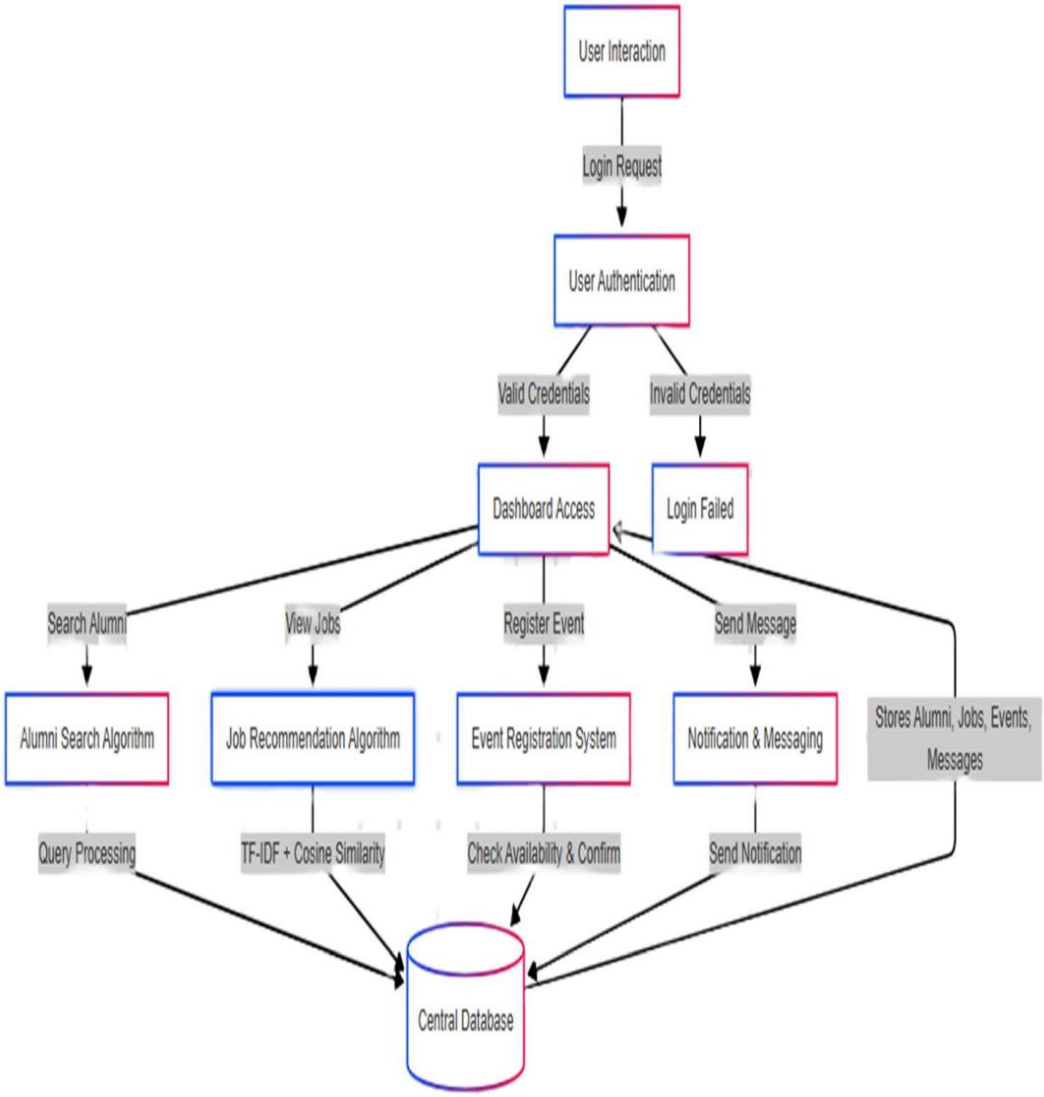

Figure 6.3 Algorithm representation

6.4 Code:

Index.html:

```
from django import template
register = template.Library()
/ / @register.filter
def index(indexable, i):
    return indexable[i]
```

Admin.py

```
from django.contrib import admin
from. models import Event
# Register your models here.
admin.site.register(Event)
```

Models.py

```
from django.db import models
from django.contrib.auth.models import User
from django.core.validators import RegexValidator
category_choices = (
    ("Enterprenuer", "Enterprenuer"),
    ("Higher Studies", "Higher Studies"),
    ("Job", "Job"),
    ("Others", "Others"),
)
role_choices = (
```

```
    ("Consultancy", "Consultancy"),
    ("Finance", "Finance"),
    ("Software Developer", "Software Developer"),
    ("Data Analyst", "Data Analyst"),
    ("Network Security", "Network Security"),
    ("Cloud Computing", "Cloud Computing"),
    ("IT project management", "IT project management"),
    ("AI and ML", "AI and ML"),
    ("others", "others"),
)
spe_choices = (
    ("Artificial Intelligence", "Artificial Intelligence"),
    ("Machine Learning", "Machine Learning"),
    ("Cloud Computing", "Cloud Computing"),
    ("Data Science", "Data Science"),
    ("Computer Vision", "Computer Vision"),
    ("Natural Language Processing", "Natural Language Processing"),
    ("Marketing", "Marketing"),
    ("Finance Systems", "Finance Systems"),
    ("IoT", "IoT"),
    ("others", "others"),
)
degree_choices = (
    ("M Tech", "M Tech"),
    ("MSc", "MSc"),
    ("MBA", "MBA"),
```

```
    ("Others", "Others"),
)
class Alumni(models.Model):
    usn = models.CharField(max_length=10)
    user = models.OneToOneField(User, on_delete=models.CASCADE,
      related_name='alumnus_details')
    name = models.CharField(max_length=100, null=True,
      validators=[RegexValidator('[+-/%@$^&*()!:;]', inverse_match=True)])
    phone = models.CharField(max_length=10, null=True)
    email = models.CharField(max_length=100, null=True)
    rv_email = models.CharField(max_length=100, null=True)
    branch = models.CharField(max_length=50, null=True)
    year_joined = models.DateField(null=True)
    year_passed = models.DateField(null=True)
    class Meta:
      db_table = 'alumni'


class Category(models.Model):
    alumnus = models.OneToOneField(Alumni, on_delete=models.CASCADE,
      null=True, blank=True)
    category = models.CharField(max_length=20, choices=category_choices,
      default='Job')
    description = models.CharField(max_length=250, null=True, blank=True)
    class Meta:
       db_table = 'category'
class Job(models.Model):
```

```
alumnus = models.OneToOneField(Alumni, on_delete=models.CASCADE,
    null=True, blank=True)
  company_name = models.CharField(max_length=100)
  role = models.CharField(max_length=50, null=True, choices=role_choices)
  salary = models.FloatField(null=True, blank=True)
  location = models.CharField(max_length=20)
  class Meta:
    db_table = 'job'


class HigherStudies(models.Model):
  alumnus = models.OneToOneField(Alumni, on_delete=models.CASCADE,
    null=True, blank=True)
  college_name = models.CharField(max_length=100)
  specialization = models.CharField(max_length=100, null=True,
    choices=spe_choices)
  degree = models.CharField(max_length=20, null=True,
    choices=degree_choices)
  location = models.CharField(max_length=20, null=True)
  year_of_grad = models.DateField(null=True)
  class Meta:
    db_table = 'higherstudies'
```

urls.py

```
from django.urls import path
from .views import (
    AlumniListView, AlumniCreateView, AlumniUploadView,
```

```
    AlumniDeleteView, AlumniPostView, AlumniSearchView,
      AlumniUpdateView
)
from base.views import IPView
from base.decorators import switch_view
from . import views
urlpatterns = [
    path('', views.AlumniListView, name='list_alumni'),
    path('new', switch_view(AlumniCreateView.as_view(), IPView.as_view(),
      IPView.as_view()), name='new_alumni'),
    path('upload', switch_view(AlumniUploadView.as_view(),
      IPView.as_view(), IPView.as_view()), name='upload_alumni'),
    path("delete/<int:pk>/", AlumniDeleteView.as_view(),
      name="delete_alumni"),
    path('posts', AlumniPostView.as_view(), name='list_alumni_post'),
    path('search', AlumniSearchView.as_view(), name='search_alumni'),
    path("edit/<int:pk>/", AlumniUpdateView.as_view(),
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    path('update', views.update, name='update'),
    path('update2', views.update2, name='update2'),
    path('update3', views.update3, name='update3'),
]
```

Manage.py

```
#!/usr/bin/env python
"""Django's command-line utility for administrative tasks."""
import os
import sys
def main():
    """Run administrative tasks."""
    os.environ.setdefault('DJANGO_SETTINGS_MODULE', 'api.settings')
    try:
        from django.core.management import execute_from_command_line
    except ImportError as exc:
        raise ImportError(
            "Couldn't import Django. Are you sure it's installed and "
            "available on your PYTHONPATH environment variable? Did you "
            "forget to activate a virtual environment?"
        ) from exc
    execute_from_command_line(sys.argv)
if __name__ == "__main__":
    main()
```

Chapter 7

SYSTEM STUDY AND TESTIG

Feasibility Study

A feasibility study evaluates the practically and viability of developing the Alimni Management System. It includes the following aspects:

- Economic feasibility
- Technical feasibility
- Operational Feasibility
- Legal Feasibility
- Schedule Feasibility

1. Technical Feasibility

Technical feasibility determines whether the existing technology can support the development and deployment of the system. This includes evaluating hardware, software, network capabilities, and development tools.

Key Aspects:

- **Programming Languages & Frameworks** – The system will be developed using Django (Python) for the backend and ReactJS for the frontend to ensure a robust and scalable platform.
- **Database Requirements** – The system will use PostgreSQL or MySQL as the primary database to store alumni records, job postings, and event details.
- **Hosting & Server Considerations** – A cloud-based solution like AWS, Azure, or Digital Ocean will be used for high availability and reliability.
- **Security Measures** – Encryption protocols, authentication mechanisms (OAuth, JWT), and SSL certificates will be implemented to protect user data.

Example:

If the alumni platform requires real-time messaging, WebSockets can be used to facilitate instant communication between alumni and the institution..

2. Economic Feasibility

This assesses the cost-effectiveness of the project by comparing development expenses with expected benefits.

Key Aspects:

- **Development Costs** – This includes the salaries of developers, testers, and system architects.
- **Operational Costs** – Maintenance, hosting, and customer support expenses.
- **Software & Licensing** – Open-source tools (like Python, Django, and PostgreSQL) will be used to minimize costs.
- **Revenue Generation** – Monetization can be achieved via premium membership plans, job postings, and event sponsorships.

Example:

Instead of purchasing expensive proprietary software, using open-source alternatives like

Django, PostgreSQL, and React significantly reduces costs.

3. Operational Feasibility

Operational feasibility assesses whether the system meets user needs and functions efficiently in a real-world scenario. It ensures that users can interact with the system without difficulties.

Key Aspects:

- **User-Friendliness** – The system should have an intuitive user interface for alumni, faculty, and administrators.
- **Automation of Processes** – The system will automate manual tasks like event registrations, job postings, and alumni verification.
- **Training & Adoption** – A small learning curve is essential for the successful adoption of the system by alumni and university staff.

Example:

If alumni find the profile update process too complicated, they may abandon the platform.

Solution: Implement a step-by-step guided form for ease of use.

4. Schedule Feasibility

Schedule feasibility evaluates whether the project can be completed within the specified time constraints without compromising quality.

Key Aspects:

- **Development Timeline** – Breaking the project into phases such as planning, development, testing, and deployment.
- **Agile Development Approach** – Iterative development with weekly sprints for rapid progress.
- **Risk Management** – Identifying potential delays (e.g., server issues, debugging challenges) and planning contingencies.

Example:

If the project has a 6-month deadline, the team should follow this timeline:

1. **Month 1-2:** Requirement gathering & UI/UX design
2. **Month 3-4:** Backend & frontend development
3. **Month 5:** System testing & debugging
4. **Month 6:** Deployment & user training

5. Legal Feasibility

This ensures that the project complies with legal regulations and data security requirements.

- **Compliance with Data Protection Laws** – The system must adhere to laws such as
- **GDPR (Europe), CCPA (California), and IT Act 2000 (India)** to protect alumni data.
- **Terms & Conditions** – Clear policies regarding data collection, usage, and user rights should be specified

- **Intellectual Property Considerations** – If using third-party APIs (e.g., LinkedIn Jobs API), licensing agreements must be reviewed.

Example:

If the platform collects alumni contact information, it must provide an opt-out mechanism to comply with GDPR regulations.

7.2 TypesofTests&TestCases

7.2.1 Unit testing

Definition:

Unit testing focuses on testing individual components or functions of the system in isolation to verify that they work correctly.

Purpose:

- Ensures that each module functions correctly.
- Ensures that each unit performs as expected in isolation.
- Helps identify bugs early in the development cycle.

Tools Used:

- PyTest (for Python-based modules)
- JUnit (for Java-based components)
- Mocha & Chai (for JavaScript applications)

7.2.2 Integration testing

Integration testing ensures that different modules of the system work correctly together when combined. It validates data flow between components.

Purpose:

- Ensures seamless communication between the User Interface, Backend, and Database.
- To verify that different modules and components work together correctly.
- Identifies issues in data exchange between different system modules.

Tools Used:

- JUnit & TestNG (for backend integration testing)
- Postman (for API communication testing)
- Selenium (for browser automation and integration testing)

7.2.3 Integration Testing

Integration testing is verifies that multiple parts of a software application work together as intended.

Purpose:

- To check how the system performs under different conditions (e.g., high traffic, large data loads).
- Ensures that the system does not crash under peak loads.
- Measures response time, latency, and scalability

Tools Used:

- Apache JMeter(for load and stress testing)
- Gatling (for simulating user load)
- Locust (Python-based performance testing tool)

7.2.4 Functional testing

Functional testing ensures that the system performs according to the specified requirements. It validates the input-output behaviour of the application.

Purpose:

- Ensures that the system meets business requirements and user expectations.
- Focuses on real-world scenarios rather than just individual components.
- Validates core functionalities like registration, login, profile management, job posting, and alumni search.

Tools Used:

- Selenium WebDriver (for UI automating browser-based functional tests)
- JMeter (for functional API testing)
- TestComplete (for UI and functional test automation)

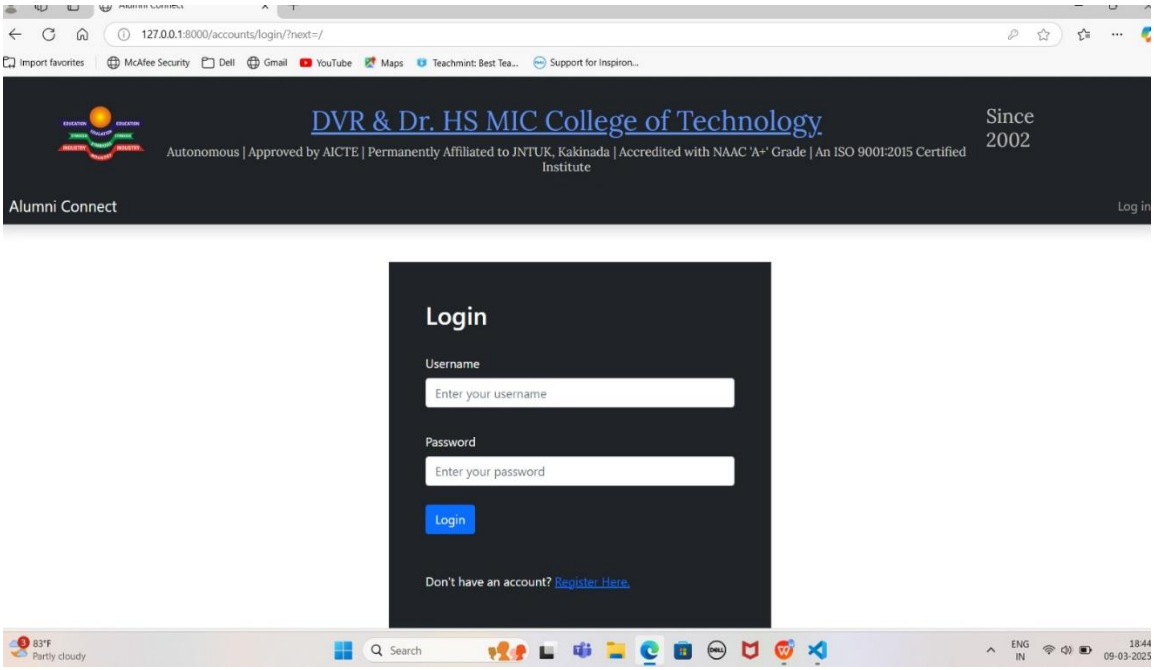
7.3 TestCases:

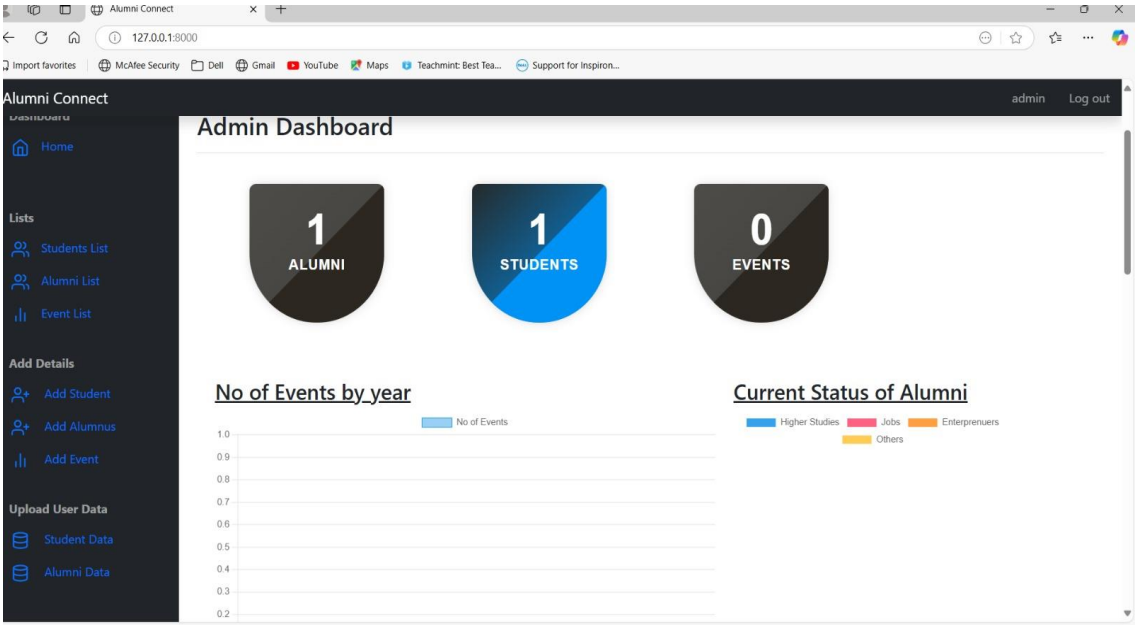
Test Case ID	Test Scenario	Test Steps	Expected Result	Actual Result	Status
TC001	User Login	1. Open login page 2. Enter valid credentials 3. Click "Login"	User is successfully logged in	As expected	Pass
TC002	Invalid Login Attempt	1. Open login page 2. Enter invalid credentials 3. Click "Login"	Display error message "Invalid credentials"	As expected	Pass
TC003	Alumni Profile Update	1. Navigate to profile 2. Edit profile details 3. Save changes	Profile details updated successfully	As expected	Pass
TC004	Event Registration	1. Select an event 2. Click "Register" 3. Confirm registration	Registration successful, confirmation email sent	As expected	Pass
TC005	Search Alumni	1. Enter search criteria (name, batch, department) 2. Click "Search"	Displays relevant alumni profiles	As expected	Pass

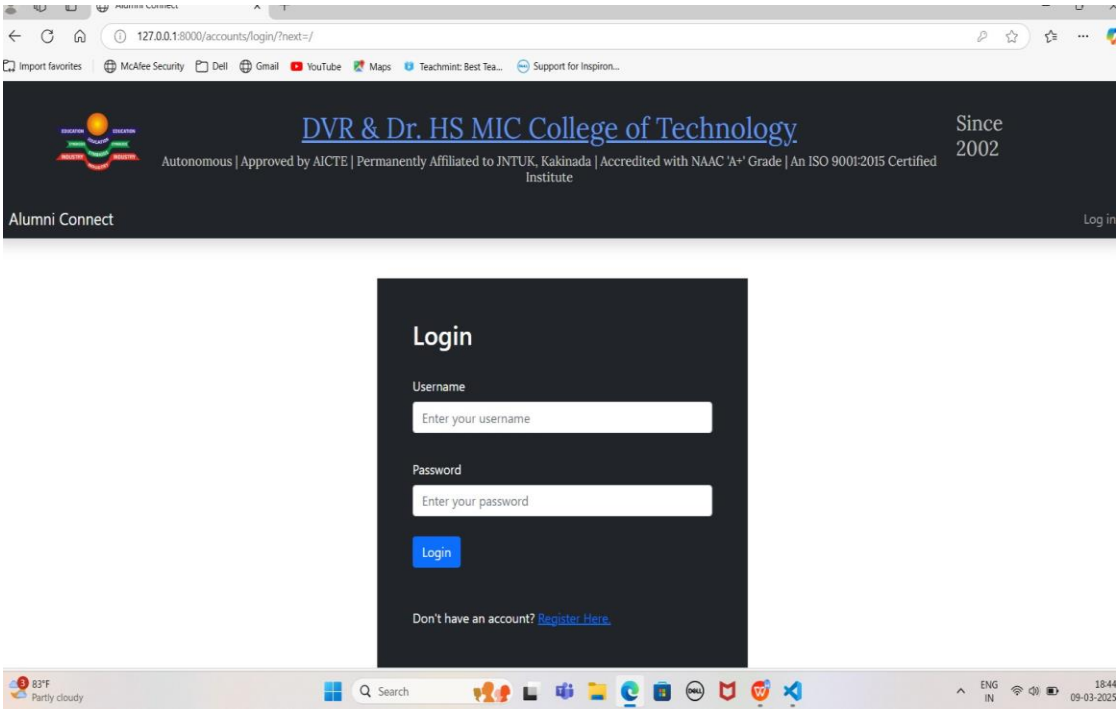
Figure 7.3 Test Cases

Chapter 8

SCREENSHOTS



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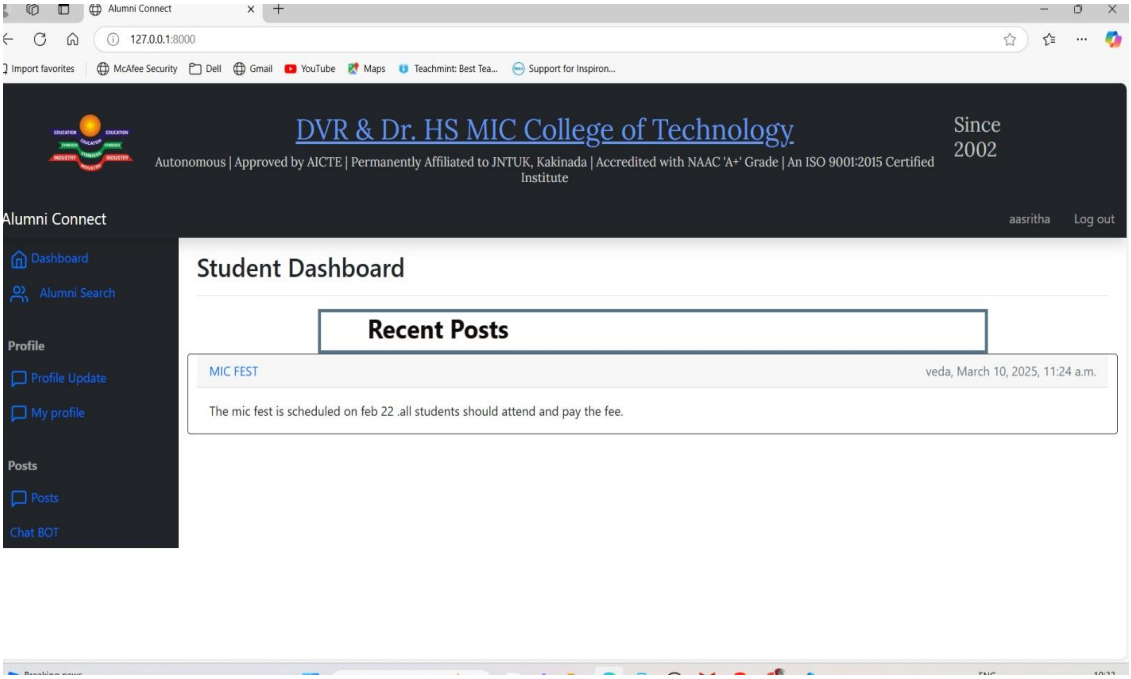
DVR & Dr. HS MIC College of Technology
Since 2002
Autonomous | Approved by AICTE | Permanently Affiliated to JNTUK, Kakinada | Accredited with NAAC 'A+' Grade | An ISO 9001:2015 Certified Institute
Alumni Connect
Log in
Login
Username
Enter your username
Password
Enter your password
Login
Don't have an account? Register Here.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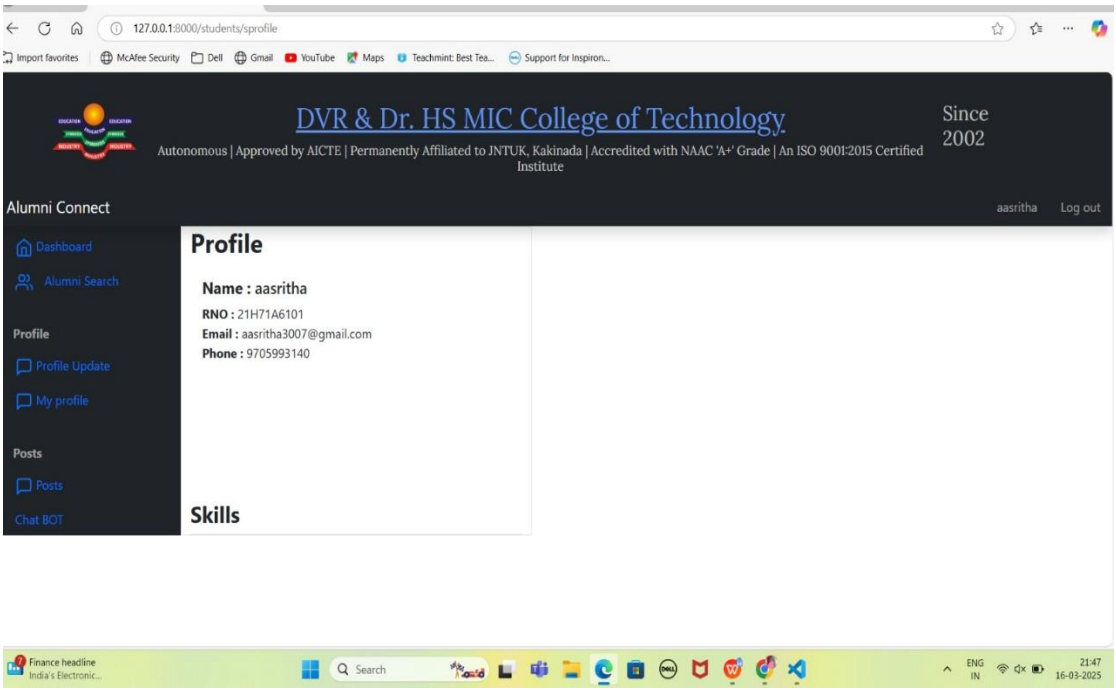
Alumni Connect
aasritha
Log out
Dashboard
Alumni Search
Profile
Profile Update
My profile
Posts
Posts
Chat BOT
Student Dashboard
Recent Posts
MIC FEST
veda, March 10, 2025, 11:24 a.m.
The mic fest is scheduled on feb 22 .all students should attend and pay the fee.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DVR & Dr. HS MIC College of Technology
Since 2002
Autonomous | Approved by AICTE | Permanently Affiliated to JNTUK, Kakinada | Accredited with NAAC 'A+' Grade | An ISO 9001:2015 Certified Institute
Alumni Connect
aasritha
Log out
Dashboard
Alumni Search
Profile
Profile Update
My profile
Posts
Posts
Chat BOT
Profile
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RNO : 21H71A6101
Email : aasritha3007@gmail.com
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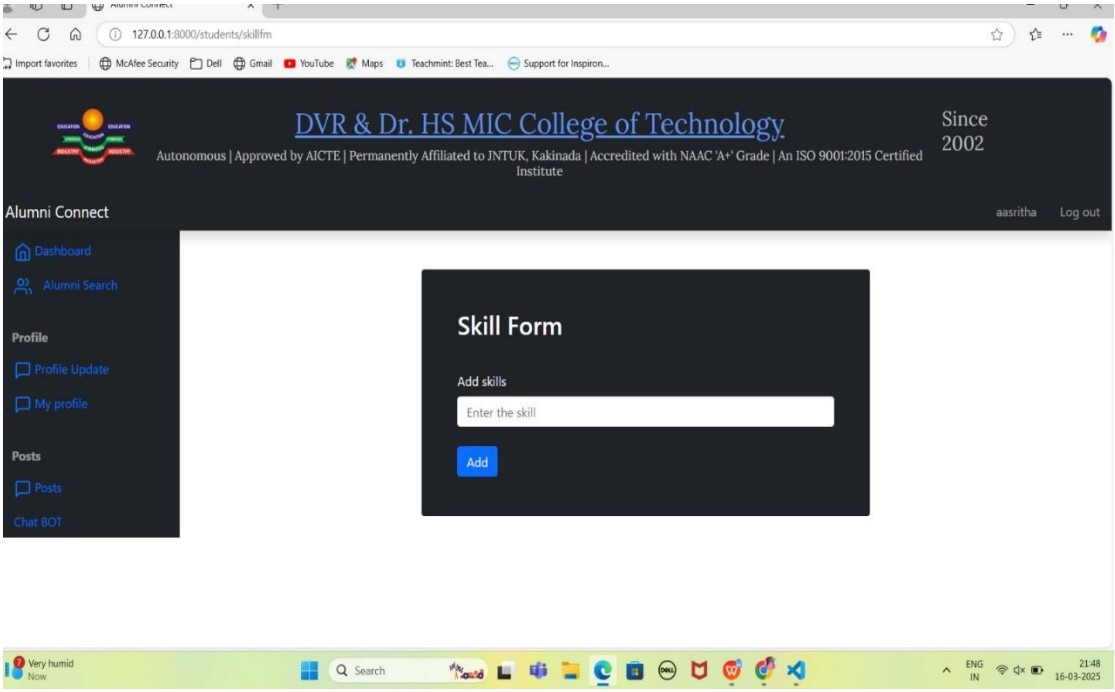
Skills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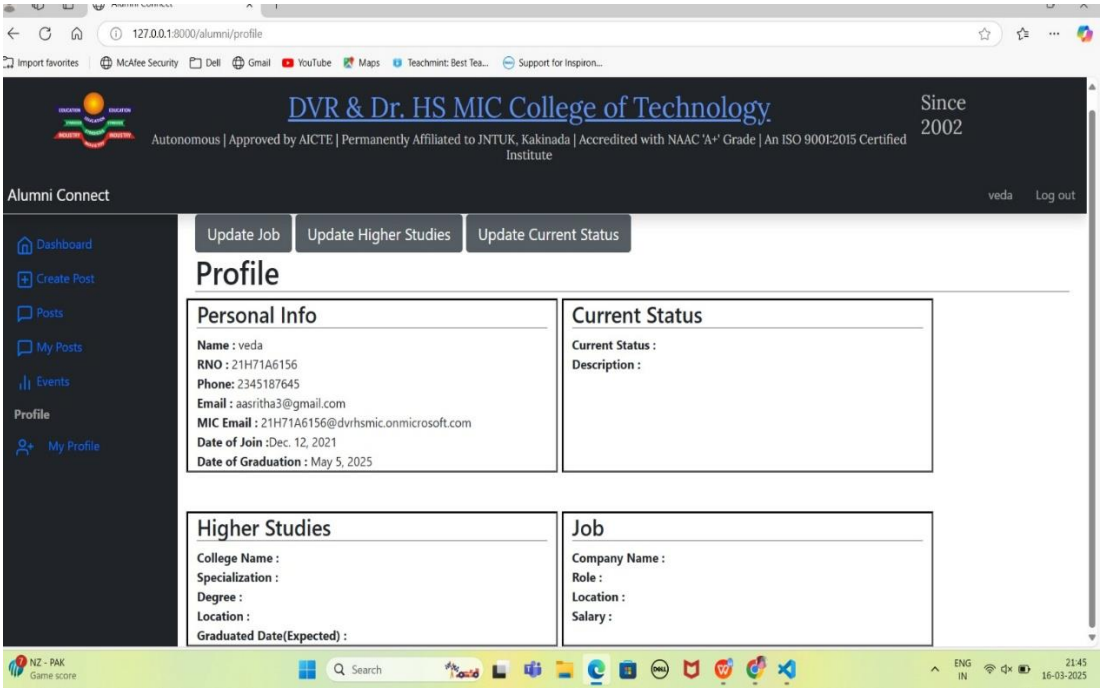
Alumni Connect
aasritha
Log out
Dashboard
Alumni Search
Profile
Profile Update
My profile
Posts
Posts
Chat BOT
Skill Form
Add skills
Enter the skill
Add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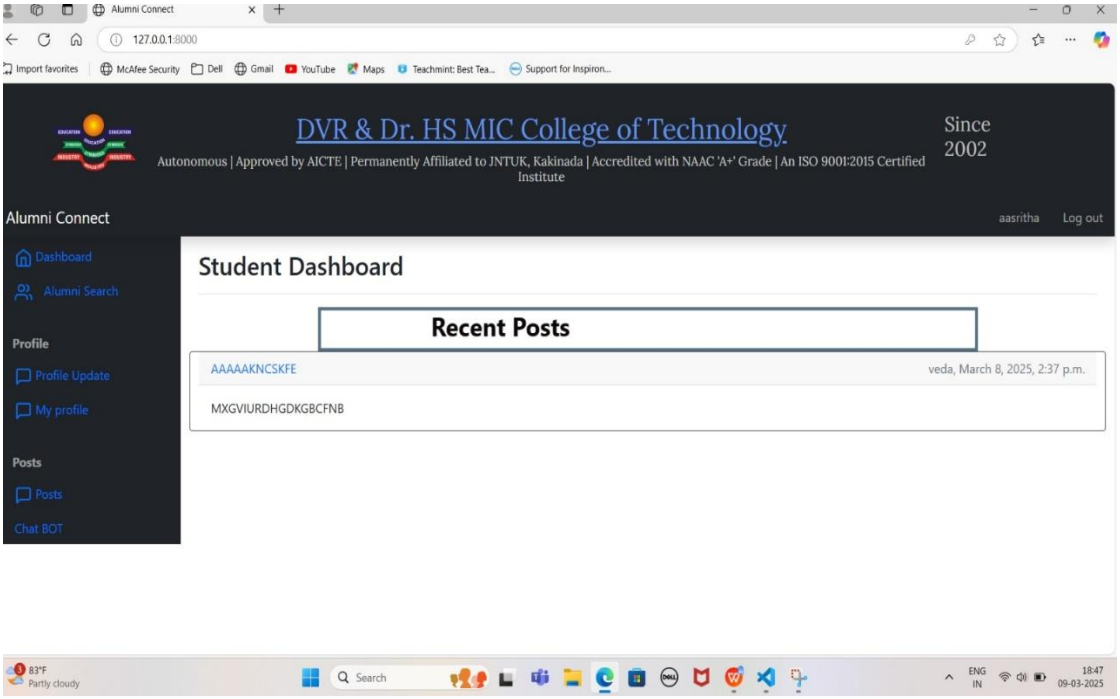
Alumni Connect
veda
Log out
Dashboard
Create Post
Posts
My Posts
Events
Profile
My Profile
Update Job
Update Higher Studies
Update Current Status
Profile
Personal Info
Name : veda
RNO : 21H71A6156
Phone: 2345187645
Email : aasritha3@gmail.com
MIC Email : 21H71A6156@dvrhsmic.onmicrosoft.com
Date of Join :Dec. 12, 2021
Date of Graduation : May 5, 2025
Current Status
Current Status :
Description :
Higher Studies
College Name :
Specialization :
Degree :
Location :
Graduated Date(Expected) :
Job
Company Name :
Role :
Location :
Salary :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aasritha
Log out
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Alumni Search
Profile
Profile Update
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Student Dashboard
Recent Posts
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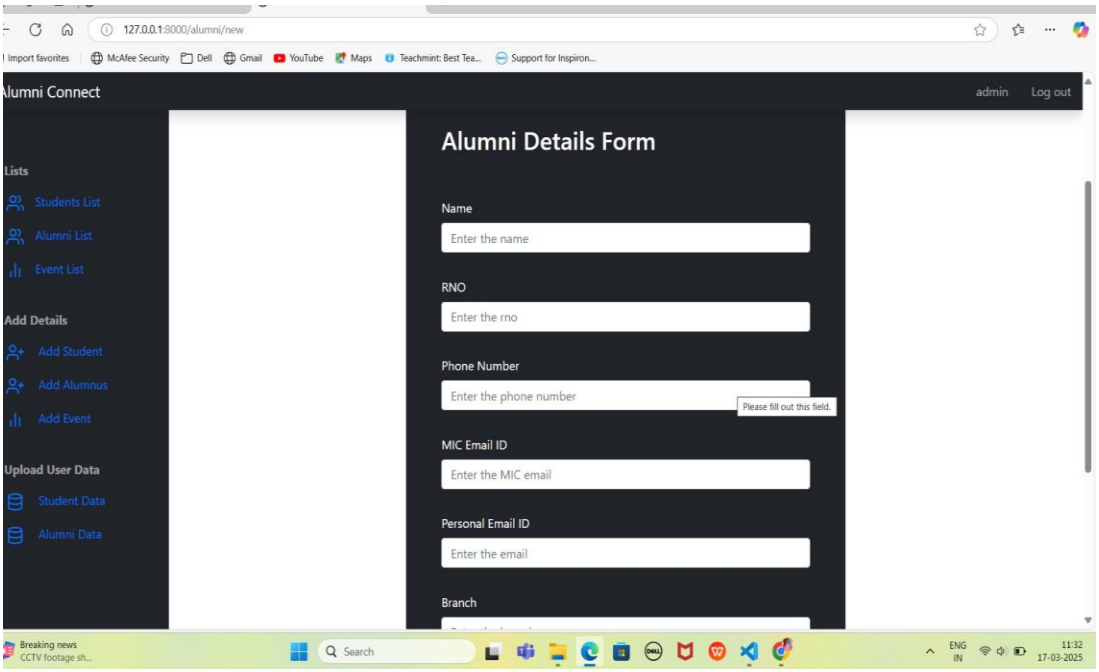
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Import favorites
McAfee Security
Dell
Gmail
YouTube
Maps
Teachmint: Best Tea...
Support for Inspiron...
Alumni Connect
admin
Log out
Lists
Students List
Alumni List
Event List
Add Details
Add Student
Add Alumnus
Add Event
Upload User Data
Student Data
Alumni Data
Alumni Details Form
Name
Enter the name
RNO
Enter the rno
Phone Number
Enter the phone number
Please fill out this field.
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Enter the MIC email
Personal Email ID
Enter the email
Branch
Breaking news
CCTV footage sh...
Search
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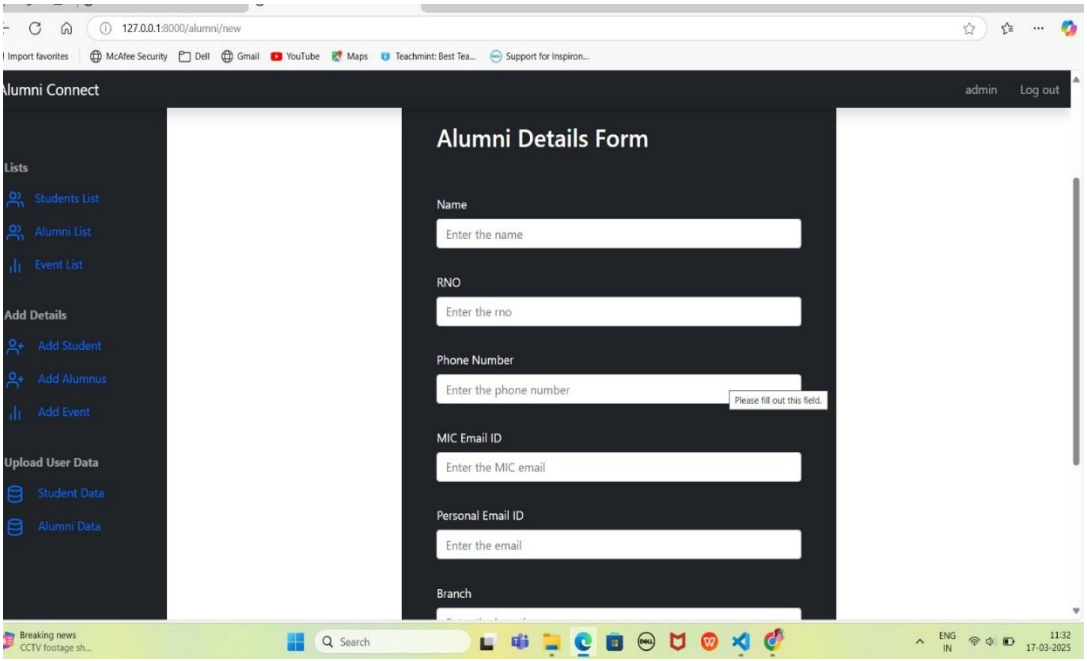
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Import favorites
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Dell
Gmail
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Support for Inspiron...
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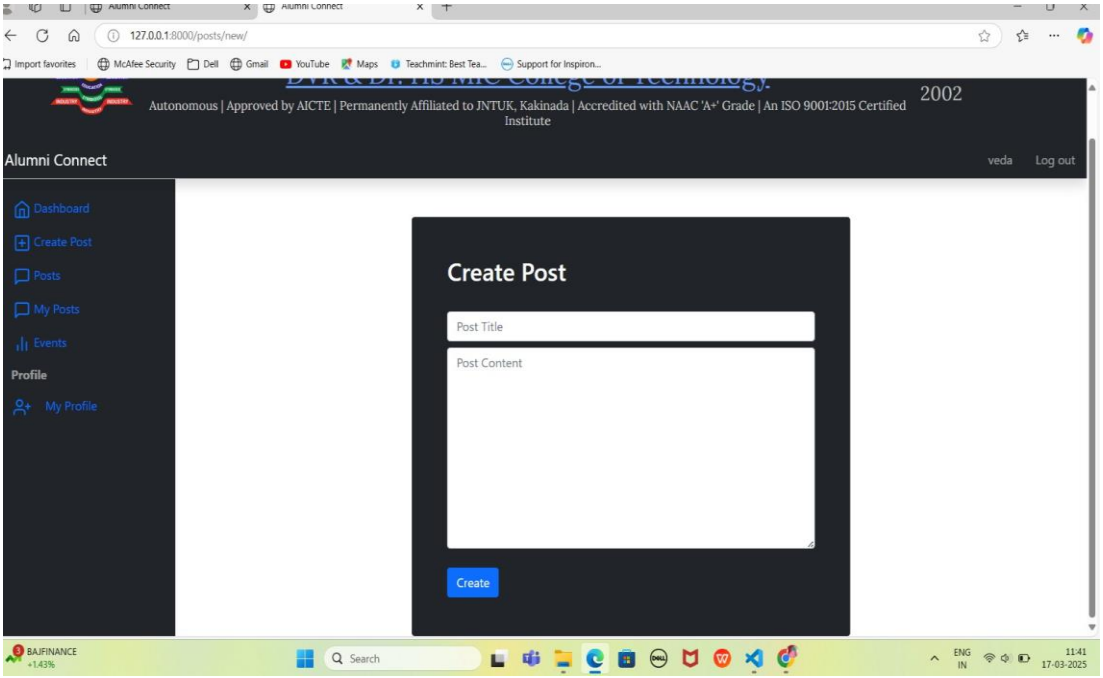
admin
Log out
Lists
Students List
Alumni List
Event List
Add Details
Add Student
Add Alumnus
Add Event
Upload User Data
Student Data
Alumni Data
Alumni Details Form
Name
Enter the name
RNO
Enter the rno
Phone Number
Enter the phone number
Please fill out this field.
MIC Email ID
Enter the MIC email
Personal Email ID
Enter the email
Branch
Breaking news
CCTV footage sh...
Search
ENG
IN
11:32
17-03-2025



URL: http://127.0.0.1:8000/

CONCLUSION

The Alumni Management System successfully streamlines and enhances communication between alumni and institutions by providing an efficient, user-friendly, and automated platform. Through this project, we have implemented a structured and scalable system that allows alumni to stay connected, share opportunities, and contribute to their alma mater.

The system's integration of frontend, backend, and database components ensures a smooth and interactive user experience. Various modules, including user authentication, alumni profiles, job postings, and event management, contribute to a comprehensive solution that meets institutional and alumni needs. The implementation of secure authentication, data handling, and real-time communication ensures the reliability and effectiveness of the system.

Through rigorous testing and analysis, the system has demonstrated its ability to handle alumni interactions efficiently. Future enhancements may include AI-driven analytics for alumni engagement, mobile application support, and blockchain-based verification for secure credential validation.

In conclusion, this project lays a strong foundation for an Alumni Management System that fosters lifelong relationships between institutions and their graduates. It serves as a digital bridge, ensuring connectivity, engagement, and opportunities for the alumni network.

Chapter 9

FUTURE ENHANCEMENTS

The Alumni Management System can be further improved with several key enhancements. Developing a mobile application for both iOS and Android will provide alumni with seamless access to the platform, real-time notifications, and event updates. Integrating AI-driven analytics can personalize user engagement by recommending job opportunities, networking connections, and relevant events based on alumni interests and activities. To enhance security and credibility, blockchain - based credential verification can be implemented to validate alumni degrees and certificates. The job portal can be upgraded with AI-powered job matching and structured mentorship programs, allowing alumni to connect with students and professionals more effectively. Additionally, integrating real-time messaging and video conferencing will improve communication within the network. Gamification features, such as a points-based rewards system, can encourage alumni to participate in events, mentorship programs, and fundraising activities. Migrating to a cloud-based infrastructure will enhance system scalability and performance, ensuring smooth operation as user engagement increases. Social media integration with platforms like LinkedIn and Facebook can facilitate better networking and outreach. Automated email and SMS notifications will keep alumni informed about important updates, while a crowd funding platform can enable secure fundraising for scholarships, institutional development, and community initiatives. These enhancements will make the Alumni Management System more interactive, efficient, and beneficial for both alumni and institutions.

Chapter 10

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